Team Syntax: Full Feature Brief for UI and Deck Teams

TIC Hackathon 3.0 · Fri 23 Oct 2026, 6 PM · Feature list v7

One-liner: A product checker that people and AI assistants use to tell whether anything is genuine, with evidence, and it gets smarter with every scan.

Pitch line: "A faker can copy a number. They can't copy a consistent history."

Product name: TBD (decide tonight; needed on the UI and slide 1).

Labels: Core = in the demo, must work by Sat 12 PM. Stretch = only if Core is done by Sat morning. Slide = pitch only.

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1. Who uses it

- **People:** scan a product with a photo or short video in our web app.
- **AI assistants:** any AI assistant (e.g. Claude desktop app) connects to our checker through MCP and asks "is this genuine?" before buying. We do not build a shopping agent and nothing is bought for real.
- Both use the same engine, and every scan makes the system smarter for everyone.

2. AI in the product

- **Research agent (Core, the one agent we build):** handles products we haven't seen. It searches official pages, genuine images, the registry, known counterfeit tells and prices, decides which checks apply, and saves what it learns. Its steps show live in the UI.
- **Single AI calls (Core):** reads the product from photos/video, compares packaging and logos with genuine references, picks the next photo to ask for, explains the verdict in plain English (Pidgin is Stretch), and turns NAFDAC alerts into a known-fakes list.
- The verdict itself comes from fixed code (checks + scoring), so the same product always gets the same answer.

3. Inputs

- Core: Photo (one or more)
- Core: Video (5–15 s); best frames picked and compressed on the phone before upload (works on slow data)
- Core: Typed details
- Core: Listing details sent by an AI assistant (images, title, price, seller)
- Stretch: Listing links (Instagram, Jumia, Jiji)

4. User situations

- Core: Before buying → Buy / Be careful / Don't buy
- Core: Already bought → Safe to use / Be careful / Don't use
- Stretch: At the store, can't open the pack (never asks to open)
- Stretch: Post-purchase follow-up that continues the same scan

5. Identification

- Core: Category detected automatically, with confidence; asks one question when unsure; user can correct it
- Core: Brand, product, model/variant
- Core: Identifiers: barcode, NAFDAC number, SON marks, batch, dates, serial/IMEI, QR; for fashion: style code, size tag, colourway
- Core: Label contents (ingredients, strength, quantity) explained in plain language
- Core: Packaging features: logo, fonts, spelling, print quality

6. Checks

Core - Barcode check digit, date sanity, spelling and print quality, details agree with each other - NAFDAC registry fit: the number belongs to this exact product; registration active - Label contents match the registration - Known fakes from NAFDAC public alerts - SON mark check: MANCAP, NIS or SON authentication mark (PAM) present on goods that need one; PAM QR decoded and user told to confirm by SMS to 281 - Clone detection: same batch seen with a different expiry; same serial in many places - Prior reports: "Reported by N other buyers in " - Visual comparison with a genuine reference, including logo checks on any product (shape, proportions, spacing, font, colour, placement)

Stretch - Fashion details: guided photos (logo, inside tag, stitching, sole/hardware, box label); style code matches tag and colourway; construction tells - Phones: IMEI format, box IMEI vs phone IMEI - Seal and tamper checks; contents visible from outside (liquid colour, fill level, tablets through windows) - Content inspection when opened, only when needed (e.g. toothpaste strip, tablet, liquid); smell/foam/texture questions - Restricted ingredients (verified list only) - Price vs market; seller red flags; seller/market reputation - Perfume batch-code decoding; batch far from its usual regions - Video hologram colour-shift; 360° seal view - Real SON database lookup (if one is found)

Honest limits (say these, never hide them) - Photos can't prove what's inside chemically. - High-quality fashion replicas can pass photo checks, so fashion verdicts cap confidence. - When unsure, the answer is "Not enough evidence" plus the next photo to take.

7. Result (what the user sees)

- Core: Verdict: Likely genuine (green) · Suspicious (amber) · Likely fake (red) · Not enough evidence (grey); always icon + text, never colour alone
- Core: Trust score (0–100) and confidence (low / medium / high)
- Core: Product identified, with "Wrong product? Edit"

- Core: Recommendation (depends on before buying / already bought)
- Core: Evidence list: each check with pass / fail / not checked, one line of detail, and source
- Core: "Not checked" line (e.g. "Contents not checked")
- Core: Contents summary
- Core: Next-step card that opens the camera ("Take a close photo of the batch number")
- Core: Report button; "Reported by N buyers" banner
- Never show "100% genuine"

8. Reporting

- Core: "Report this product" on any suspicious or fake result, bought or not
- Core: Saves product, batch, dates, photos, verdict, optional seller/shop name, state and market
- Core: Anyone who scans the same batch later sees "Reported by N other buyers"
- Core: AI assistants can report too (MCP `report_suspect`)
- Stretch: Shareable evidence report for refunds; prefilled NAFDAC report (0800-162-3322, sf.alert@nafdac.gov.ng); seller/market reputation

9. AI-assistant connection (MCP)

- Core: `verify_product` (photo, video, listing or typed details → verdict)
- Core: `report_suspect`
- Core: Demo: an existing AI assistant compares two listings using our checker
- Stretch: `get_risk_map`, `lookup_reference`

10. Memory (gets smarter with every scan)

- Core: Every scan saved (batch, dates, serial, location) → powers clone detection
- Core: Reports saved → powers "reported by N buyers"
- Core: Everything the research agent learns is saved, so the next scan of that product is instant
- Core: Demo items pre-loaded

11. Screens

Core 1. **Scan:** Take photo · Record video · Type details; "Already bought this?" toggle 2. **Checking:** live steps (Identifying product → Reading NAFDAC number → Checking registry → Checking scan history → Researching product) 3. **Result:** everything in section 7 4. **Report:** prefilled product; seller (optional); state + market; confirm → "Anyone who scans this batch will now see your report."

Stretch - Scan history · Dashboard (scan log, clone alerts, reports, heatmap) · Pidgin toggle · Telegram/WhatsApp bot · evidence report - Fashion guided-photo screens · content inspection screens

Demo states to mock now (before the backend is ready) 1. Genuine drug pack → likely genuine; next step = batch photo 2. Registry mismatch → likely fake: "NAFDAC number belongs to a different product" 3. Clone → likely fake: "Batch seen with a different expiry in Kano" 4. Reported batch → suspicious + "Reported by 1 other buyer" 5. SON item with mark → evidence row "SON mark present; confirm by SMS to 281" 6. Unknown item → research steps streaming, then verdict or "Not enough evidence" + next photo

12. Demo (2 minutes)

1. A person scans a real pack (photo or video) → verdict with evidence
2. Registry mismatch: the number belongs to another product
3. Clone detected: batch seen with a different expiry
4. Scan a fake → Report → scan the same batch again → "Reported by another buyer"
5. An AI assistant compares two listings using our checker
6. "Hand us anything": live research on a judge's item (if time; otherwise in pre-judging or Q&A)

13. Pitch deck (PDF only, max 15 slides, done by Sat 12 PM)

1. Title: name, one-liner, team
2. Hook: Feb 2026, NAFDAC seized over 10 million doses of fake malaria drugs and cosmetics worth ₦3bn at Lagos Trade Fair Market + one survey story
3. Problem: fakes everywhere and impossible to check; fakers copy real NAFDAC numbers (fake BETACLOX carried a number belonging to Mebendazole 500mg)
4. Why current fixes fail: copied numbers beat lookups; scratch codes cover few brands; raids come after the sale; nothing for AI assistants
5. Survey: "We surveyed X people here last night: Y% bought a fake this year"
6. Why now: AI assistants are starting to shop for people and need to check too
7. Solution: one checker for people (photo/video) and AI assistants (MCP); any product
8. How it works: evidence stacking: NAFDAC registry fit, SON marks, clone detection, logo/visual checks, known fakes, buyer reports
9. Demo
10. AI at the core: research agent, memory, MCP
11. Gets smarter with every scan: scan history, clones, reports
12. Business model: API fees (marketplaces, pharmacies, AI-assistant platforms); brand subscriptions (signed codes + counterfeit intelligence); free consumer app
13. Nigeria first → Ghana, Kenya, South Africa (country adapters; AfCFTA)
14. What we built in 24h + what's next (live NAFDAC/SON integrations, brand partnerships, signed QR, more countries, offline mode)

Stats (verify before final, cite on slide) - ₦3bn Trade Fair seizure (Feb 2026): allafrica.com/stories/202602110135.html - WHO: ~1 in 10 medical products in low- and middle-income countries substandard or falsified (2017): tralac.org/news/article/12471 - INTA Gen Z Nigeria: 97% bought counterfeits in the past year (check year): inta.org (Gen Z Insights Nigeria report) - BETACLOX copied number: channelstv.com/?p=977209 - Postinor2 fakes reusing the genuine registration number: gazettengr.com/?p=402521

Do not claim - 100% detection, or that we "authenticate" anything - Live NAFDAC or SON integration (we use public data and on-pack marks) - Lab-grade content testing - Any number we haven't verified

Pre-judging answers (one sentence each) 1. Problem: fakes are everywhere in Nigeria and impossible to check; buyers, and the AI assistants shopping for them, can't tell if a product is genuine before they pay. 2. Users: anyone buying in Nigeria, plus AI assistants, marketplaces and pharmacies buying for them. 3. What it does now: [fill Sat morning] 4. What we built in 24h: [fill Sat morning] 5. Next: live regulator integrations, brand partnerships for signed codes, more countries.

Tonight: survey (~40 people) by 9 PM · pick the product name · draft slides 1–8 and 11–14 · backup video Sat morning.